BVRIT HYDERABAD
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Department of Electronics and Communications Engineering

SKIN CANCER DETECTION

TEAM:7

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AGENDA

- Problem Statement
- Dataset
- Python Packages and tools
- Modules
- Implementation
- Algorithm
- Team contribution
- References
- Execution on final code

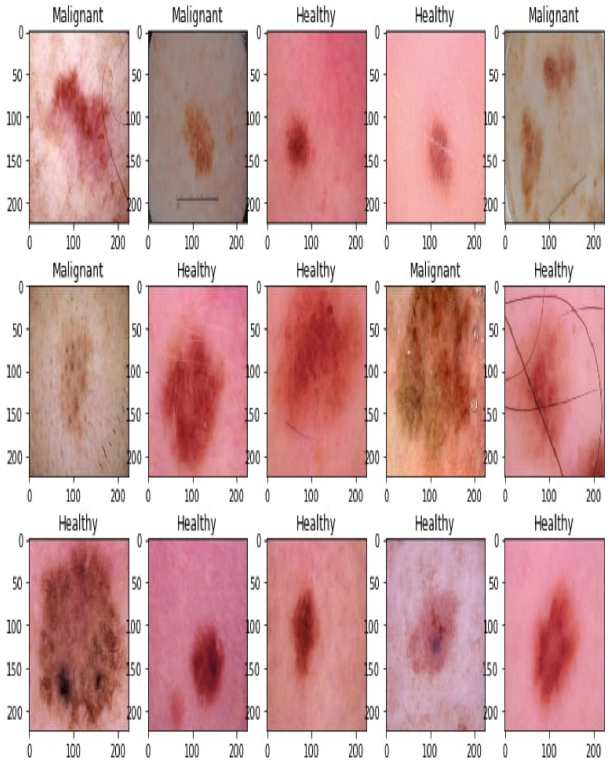
PROBLEM STATEMENT

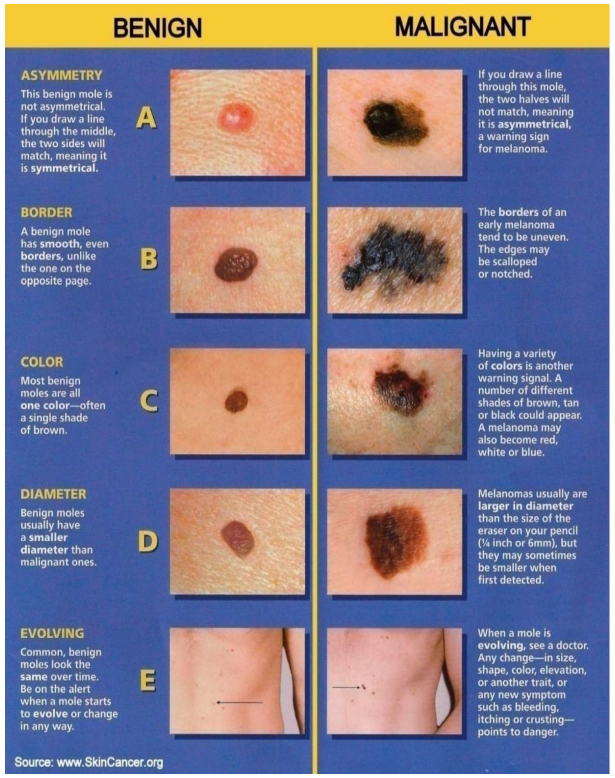
- Skin cancer is one of the most dangerous forms of cancer. Skin cancer is caused by unrepaired deoxyribonucleic acid (DNA) in skin cells, which generates genetic defects or mutations on the skin.
- Skin cancer tends to gradually spread over other body parts, so it is more curable in the initial stages, which is best detected early.
- Lesion parameters such as symmetry, color, size, shape, etc. are used to detect skin cancer and to distinguish benign skin cancer from melanoma.
- These days technology is rapidly widespread so using Machine Learning we preprocess the image. We analyze the image with trained data and we predict the Cancer. The proposed method detects melanoma faster than the biopsy method.

DATASET

- Dataset used for this are extracted from kaggale towards skin cancer detection
- It consists of 3000 images of skin cancer
- The training data consists of 2000 images and testing data consists of 1000 images

IMPLEMENTATION





PYTHON PACKAGES AND TOOLS

PACKAGES	TOOLS
TensorFlow	unzip
NumPy	
Keras	

PACKAGES

- Tensorflow: Tensorflow is used for data automation, model tracking, performance monitoring
- NumPy: NumPy is used to perform a wide variety of mathematical operations on arrays
- Keras: Keras is a high-level,deep learning it is used for implementing neural networks

ALGORITHM

- A convolutional neural network (CNN or ConvNet) is a network architecture for deep learning that learns directly from data.
- CNNs are particularly useful for finding patterns in images to recognize objects, classes, and categories.
- Accuracy of final model is 82 percentage

MODULES

- os: The os module is used for interacting with the operating system, setting environment variables, and managing file paths.
- Pandas: Pandas is a data manipulation library used for working with structured data, such as data from CSV files.
- PIL: PIL (Python Imaging Library) is used for image processing. It is often used for opening, manipulating, and saving image files

FRAMEWORK OF DEPLOYMENT

- Streamlit is a Python library that is well-suited for building interactive web applications for various purposes, including deep learning

CONTRIBUTION

NAME	ROLL NO	CONTRIBUTION
B.Leelavanitha	21WH1A04A0	Basic code(Model Building),PPT using Latex
M.LAKSHMI PRIYA	21WH1A04A1	Basic code,Error finding
P.UMA MA-HESHWARI	21WH1A04A2	Deployment using Streamlit and error finding,code
M.SIRIVALLY	21WH1A04A3	Basic code (Exploratary Data Analysis)
B.RAKSHITHA	21WH1A04A4	Basic code(Model Building),PPT using Latex

REFERENCES

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- https://ieeexplore.ieee.org/document/9696874
- https://www.ncbi.nlm.nih.gov/pmc/articles/PMC8705277/

THANK YOU